

Cybersecurity Risks and Threats in Avionics and Autonomous Systems

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Abstract—Ongoing advances in digitization and automation of critical infrastructures, particularly in terms of aeronautical Communication, Navigation, and Surveillance (CNS) technologies, has led to the fusion of intricate physical and information networks with artificial intelligence (AI) systems. This integration is empowered avionics and Air Traffic Management (ATM) systems with enhanced data processing capabilities, interactive information exchange, and expansive geographic distribution. However, these advancements also expose the systems to increasing cybersecurity threats, physical vulnerabilities, and data integrity risks. The complexity and interconnectedness of CNS infrastructure, combined with ATM systems, amplify the potential scope and depth of attacks and increase their ability to spread through interconnected components. As a result, both ATM and UAS Traffic Management (UTM) systems face an escalation in security challenges. While the concept of cybersecurity within aviation has a long history, its seamless integration into aviation systems remains a significant challenge. Just as AI technology is harnessed to improve the overall operational efficiency and reliability of aviation systems, it has also emerged as a pivotal battleground for cybersecurity risks and threats. Increasingly, AI-driven intrusion and theft methods are replacing traditional approaches. In response, researchers have put forth defensive strategies rooted in AI technology. This paper critically evaluates cybersecurity vulnerabilities and threats that may confront ATM and UTM systems. It systematically categorizes a variety of potential threat actors along with their corresponding targets, based on their objectives, motivations, and capabilities. Simultaneously, the paper delves deeply into an exploration of possible attack methodologies founded on AI technology, accompanied by their corresponding defensive tactics.

Keywords—Cybersecurity, Avionics system, Autonomous Systems, Cyber-Threat, ATM, UTM, CNS+A

I. INTRODUCTION

In 2021, the European Air Traffic Management Computer Emergency Response Team (EATM-CERT) [1] pointed out that the global aviation industry is facing cyber-attacks, and security risks are rising. The report pointed out that among all the cyber-attacks detected in 2020, fraudulent websites alone caused losses of 1 billion US dollars. In the 2023 report [2], EATM-CERT reveals at least 2.5 aviation-related attack reports per week, from ransomware alone. Most of these incidents affected original equipment manufacturers (OEMs), followed by airspace users. The main drivers of these attacks were economically related. Responding to the escalation in air traffic density, along with the drive for transparency, reliability, and promptness in information dissemination as championed by the global integration of airspace [3], coupled with the emergence of the Aviation 4.0 paradigm [4], pivotal facets within the Air Traffic Management (ATM) system are being progressively modernized. The new generation of ATM is based on the hybrid framework of CNS+A, that is, the

traditional ATM system tightly coupled with the communication, navigation, and surveillance (CNS) framework. This transition requires a transition from traditional terrestrial analogue signals to sophisticated intelligent digital systems, augmented further by space-based signals [5]. For this reason, in the planning of the next generation air transportation system (NEXTGEN) [6] in Europe and the United States, the strategic framework of the entire air transportation system is built around digital data networks. System-wide information management (SWIM) [7] assumed the backbone of NextGen's digital data sharing. The system widely adopts IP-based and wireless transmission technology, which makes real-time data sharing between aircraft and aircraft, aircraft and facilities, and facilities and facilities possible [8]. Furthermore, the surge in Unmanned Aircraft System (UAS) utilization, particularly the advent of UAS-centric commercial delivery [9], sets the stage for a proliferation of both manned and unmanned aerial platforms operating with in lower -altitude airspace. In anticipation of the anticipated rise of UAS, distinct UAS Traffic Management (UTM) [10] and U-Space [11] initiatives have been launched by NASA and Europe, respectively. These initiatives are geared to the management of UAS operations, accommodating Urban Air Mobility (UAM) services, and surmount the challenges associated with high-density activities within lower-altitude airspace. Whether it is CNS+A or future UAM services, artificial intelligence (AI) algorithms and autonomous navigation technology are indispensable cornerstones. The use of AI technology to improve the efficiency of airspace operations has been extensively studied, and great progress has been made [12-15]. However, the current ATM system has very limited defence against network intrusion of communication and surveillance systems [16]. UTM, which relies on digital avionics systems and trajectory-based operations (TBO), both in turn relying on data interaction, will face increasing cybersecurity vulnerabilities. The evolution of these threats has shifted from traditional avenues of intrusion to harness the power of AI approaches. As AI technologies continue to grow in aviation, the complexity and interdependence of CNS+A systems will inevitably grow further, expanding the attack surface. The resiliency and integrity of ATM and UTM systems and procedures need to be enhanced to deal with the large number of cyber-attacks on the horizon [17]. The purpose of this paper is to provide the ATM research and development (R&D) community with an updated and comprehensive understanding of the pervasive cybersecurity risks in ATM and UTM systems, while advancing UAM and UAM security integration. The main part of this document consists of four sections. The first part analyses and classifies the offensive strategies. The second part identifies possible threat actors based on goals, motives, and capabilities. The third section explores potential targets of cyber-attacks within ATM and

UTM systems. The fourth part is the mitigation measures based on AI technology.

II. CLASSIFICATION OF CYBER-ATTACK

Within the realm of aviation, cyberattacks assume the form of informatics breaches, with the potential to affect an array of elements within the ATM and UTM infrastructure [18]. These attacks fundamentally constitute data intrusions, manifesting through data manipulation. As an avenue of intervention, network-based attacks are wielded as mechanisms to impact both software systems and tangible hardware components. Therefore, we assume data as the core and divide the attack methods into three categories. Figure 1 illustrates three types of cyber-attacks: data interception, data manipulation, and data interference.

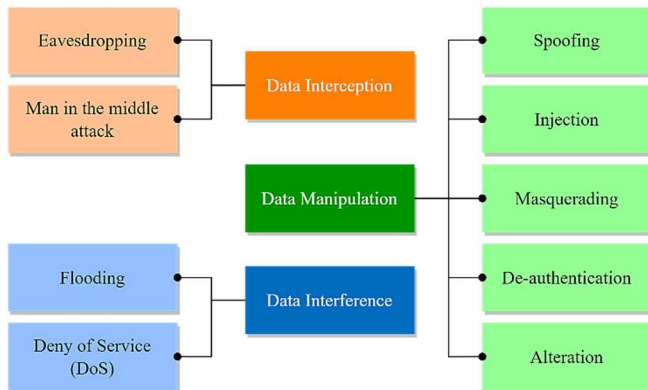


Figure 1 Types of cyber-attacks.

Data interception entails the unauthorized encroachment into repositories housing sensitive information. Often, the effectiveness of data interception is rooted in the absence of robust encryption and authentication mechanisms within system architectures and communication protocols. Unscrupulous entities exploit these vulnerabilities to skirt the authorization protocols of both ATM and UTM systems, introducing Trojan horses and viruses as vehicles to surprisingly commandeer data or privileges resident within the system [19]. Typical data interception methods include communication and data eavesdropping. The motives behind data interception span two distinct objectives: straightforward data pilferage and preliminary steps to facilitate composite attacks. In the latter scenario, malevolent actors frequently opt for a man-in-the-middle (MITM) stratagem. This entails the unauthorized interception of multi-party communications, enabling the eventual acquisition of Sensitive information and preparing the terrain for ensuing attacks.

Data manipulation is a disaster for any virtual system and physical network. In civil aviation, systems and networks between departments and stakeholders are strongly interlinked. When these key points become potential entry points, the key elements within this framework are all very vulnerable. Attackers can often exploit vulnerabilities in systems and communication protocols to infiltrate, modify, and delete data, thereby compromising data integrity. This approach and strategy emphasizes manipulation and rewriting of data by means including but not limited to MITM, GPS spoofing, ADS-B message manipulation, SQL and NoSQL injection attacks, and de-authentication operations [20-23].

Data interference is an attack strategy that requires flooding communication channels and data paths with a large amount of false data and signals, thereby causing obstacles to

data reception and transmission. A typical manifestation of this situation is a denial of service (DoS) attack, which is a method of rendering a particular service inaccessible by overwhelming the computing power of a device or the bandwidth granted to a user by an institution. For example, interference devices subtly impair the ability of avionics systems to intercept GNSS signals [24]. The result was a loss of navigation services, causing the unmanned aircraft system (UAS) to lose its ability to determine position and trajectory. The severity of data interference looms over the functionality of UTM systems. The network outage drew outcry as the UTM system dropped ties to the UAS units under its jurisdiction, leading to a massive mission collapse and hastening the loss of airspace governance.

III. SECURITY THREAT AGENTS

The proliferation of intelligent algorithms within the realm of the ATM system, combined with the forthcoming advancements in Urban Air Mobility (UAM), is poised to usher in an array of novel threats and complexities. A diverse array of malefactors stands ready to jeopardize the security of both the ATM and UTM systems. The work by Lykou [8] undertook an analysis and taxonomy of those posing cybersecurity threats to civil aviation. Concomitant parallels can be drawn between the cyber hazards confronted by ATM and UTM. Figure 2 is a classification of these threat agents. Leveraging the operational nuances distinguishing manned and unmanned aircraft, alongside the extent of their potential impact on the system and the environment, these malevolent actors can be dichotomized into three categories: low-risk pilferers, medium-risk transgressors, and high-level menace.

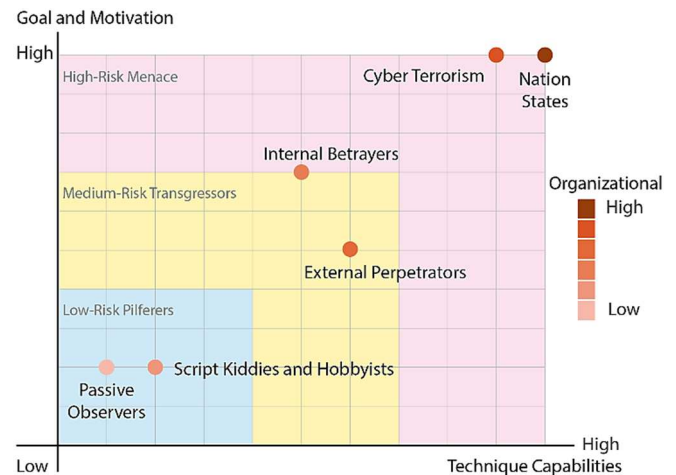


Figure 2 Classification of threat agents.

Within the realm of low-risk pilferers, their actions tend to yield negligible direct detriments to the system and its surrounding milieu. Their preferred modus operandi often revolves around data interception. Categorized as passive observers, they subtly collect data through unobtrusive surveillance. Driven by inclinations such as pastimes, personal curiosities, and an appetite for thrill, their pursuits largely orbit around benign intentions. Script Kiddies and Hobbyists adopt a modus operandi that frequently capitalizes on well-documented security vulnerabilities within systems to extract data. Predominantly composed of hobbyists and novices in the realm of hacking, their approaches typically adhere to a solitary technique [25].

Medium-risk transgressors may cause significant material damage. This category can be divided into internal defectors

and external perpetrators. Insider traitors often have the technical prowess, or the necessary authorization within the system, to easily access critical applications or bypass defensive networks to steal data. This group includes all stakeholders, and their motives may include dissatisfaction, malice or purely monetary goals. The intricacies of their internal identities make their detection a formidable challenge. Cybercriminals make up the bulk of external criminals. They identify and exploit security gaps in ATM systems and related infrastructure through techniques such as eavesdropping, signal jamming, and expropriation [26]. The primary goal of such groups is to achieve financial gain through data theft. Usually, they have ample resources and strong skills to back them up. Its routine practice is to execute ransom-based coercion and extortion by creating a credible threat by inflicting maximum damage. It is foreseeable that such groups will be the main body of future network security threats.

High-level menace is extremely dangerous. Such groups have greatly endangered national security and social balance, causing huge property and personnel losses. Cyber terrorists, extremists, and insurgents have integrated sophisticated cyber technologies due to the rapid development of civilian and military computer technology. They take advantage of the vulnerabilities of the information aviation system and the inherent vulnerabilities of the infrastructure to undermine the security of the system or pose a threat to the security characteristics of the information aviation system resources [27]. Their purpose is to disable and overrun the operation of

ATMs and avionics. Their tactics are characterized by the use of drones for intelligence reconnaissance, dissemination of extremist propaganda, and even aerial bombing [28, 29]. Its reach can extend into urban areas, creating chaos by commandeering drones to plan crashes in densely populated areas. Mwiki [30] emphasizes that state-sponsored infiltration has long been an ongoing challenge, and he carefully outlines the currently documented state-sponsored and fostered hacker collectives. Such criminals have virtually limitless resources, an in-depth knowledge of ATM and UTM systems, and a deep knowledge of a variety of avionics. This keen insight allows them to expertly circumvent redundant defence mechanisms through tactics such as data interception, communication network disruption, and infrastructure destruction, all to further their military and political ambitions.

IV. CYBERSECURITY WEAKNESSES IN CNS+A

Virtually every system and module encompassed within the CNS+A framework is susceptible to potential attacks. Blasch [31] delved into the security threats arising from vulnerabilities inherent in avionics technology, particularly their implications for the ATM system. This study dissected, evaluated, and appraised the avionics equipment that harbors latent hazards. Figure 3 provides a visual depiction of the interconnected infrastructures within the purview of the ATM and UTM systems, embedded within the overarching CNS+A framework. The subsequent sections discuss these aspects sequentially.

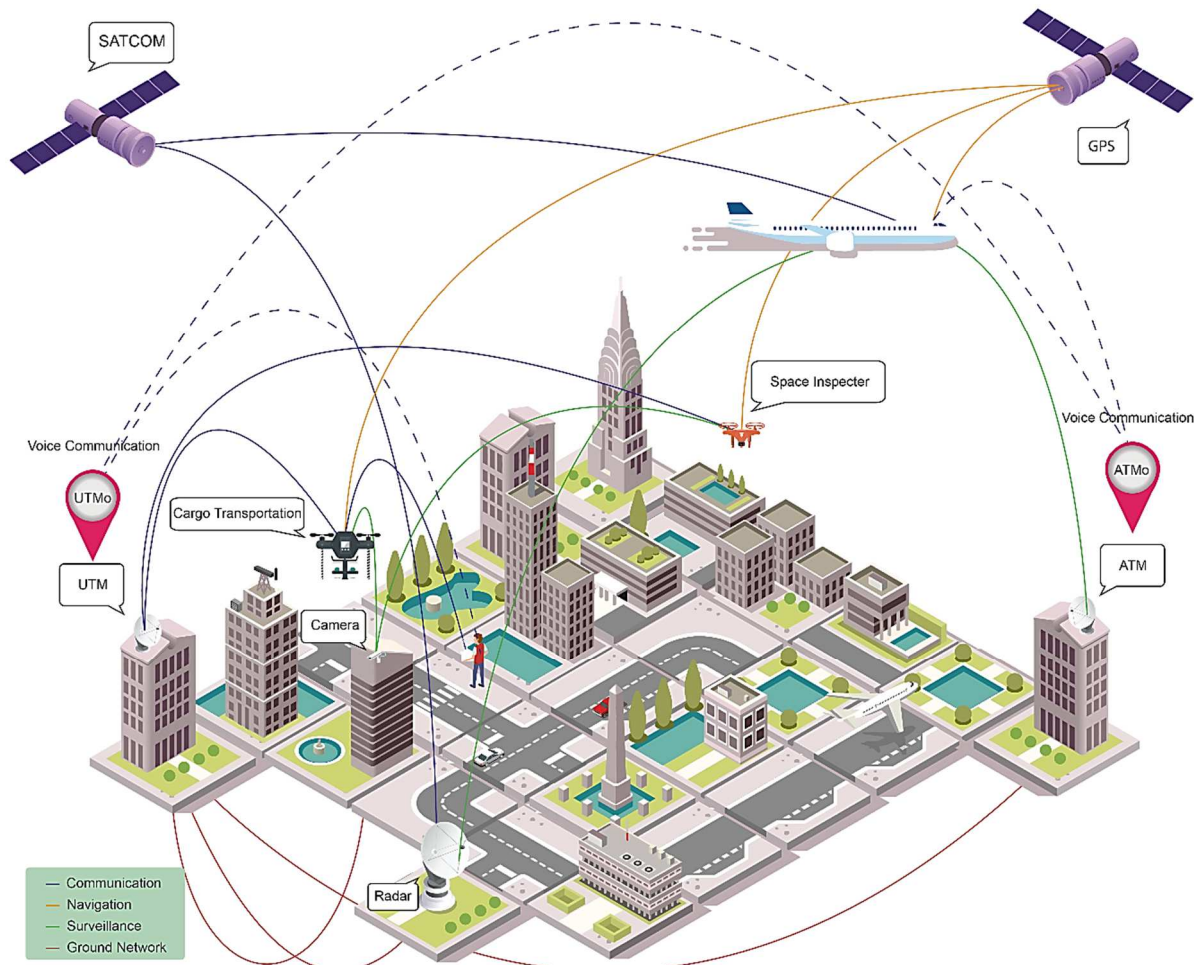


Figure 3 ATM and UTM system infrastructure in the CNS+A framework.

A. Communication

Air-ground communications are a critical means for ensuring the safe operation of aviation. Contemporary air-ground communications are rooted in the fabric of ATM operations, with multiple communication protocols, each with specific functions for interacting with airborne vehicles [32]. High frequency/very high frequency voice communications (HF/VHF) rely on analogue, unencrypted radio exchanges, which makes them vulnerable to eavesdropping and signal jamming. Controller-Pilot Data Link Communication (CPDLC) allows air traffic controllers (ATCo) and pilots to interact via a data link. Despite its digital underpinnings, the CPDLC's communication chain lacks encryption and authentication, leading to eavesdropping on data traffic that can be surreptitiously monitored by third parties without the consent of the communicating parties. At the same time, attackers can launch jamming and flooding attacks that prevent aircraft from getting service [33]. Aircraft Communications Addressing and Reporting System (ACARS) is responsible for transmitting messages in clear text over wireless channels, and thus is vulnerable to eavesdropping attacks, eventually leading to potential information leakage [34]. The security posture of L-band Digital Aeronautical Communication System (L-DACS), is further compromised due to the lack of hardware protection mechanisms, making it prone to information leakage, denial of service, and unauthorized system access [35]. For satellite communications, the most important cyber threats usually manifest as unavailability issues, typical examples are jamming attacks and targeted denial-of-service (DoS) attacks [36]. In the field of civilian drone communication, people are more inclined to use wireless networks operating at 2.4GHz and 5GHz, including technologies such as Bluetooth and Wi-Fi 802.11 [37]. However, these communication methods often lack encryption and authentication, making them subject to passive eavesdropping, active jamming, and possibly hijacking.

B. Navigation

Currently, navigation systems mainly rely on Terrestrial Radio Navigation Aids (TRNAs) and Global Navigation Satellite System (GNSS). Within TRNA, two prominent systems stand out: Instrument Landing System (ILS) and VHF Omnidirectional Range (VOR). The ILS acts as a radio-based guidance for aircraft landing, providing horizontal and vertical guidance. However, it is vulnerable to shadow attacks and mono manipulation [38]. The design of VOR deliberately omits the confidentiality feature to minimize computational overhead, so it is vulnerable to interference and spoofing [39]. The cornerstone of modern navigation systems is the Global Navigation Satellite System (GNSS), which is free from the constraints of traditional terrestrial navigation equipment. GNSS provides real-time three-dimensional position coordinates for aircraft and UAS. However, GNSS is not without security risks. GPS signals are susceptible to interference from environmental factors and jamming activities [40]. In the jamming spectrum, GPS spoofing is a technically complex endeavour with a lower success rate than GPS jamming. While GNSS-equipped aircraft can use data from TRNA devices to identify spoofing attempts in GNSS systems, the dilemma is even more dire for many civilian drones. Civilian drones, which typically rely solely on GPS for navigation, are more vulnerable to the harmful effects of GPS spoofing attacks than traditional aircraft. If a GPS

spoofing attack is successful, the drone could fall victim to capture and remote control, ultimately leading to a catastrophic crash [41].

C. Surveillance

The task of aerial surveillance in the field of civil aviation is mainly to track the flight path and altitude of the aircraft in real time, and predict potential collisions and dangers. Primary Surveillance Radar (PSR) relies on the principle of signal reflections to calculate the distance and position of an aircraft. While radar signals are immune to message injection attacks, they can be jammed [42]. Secondary Surveillance Radar (SSR) constitutes a coordinated system involving radar sensors and aircraft transponders. However, due to the unencrypted nature of the signal and the limitation of computing power of civil aviation aircraft, SSR is vulnerable to interception and tampering [43]. Automatic Dependent Surveillance-Broadcast (ADS-B), with its refresh rate of two per second and identification capability of five per second, is a powerful addition to the surveillance framework. Its scope of application extends to general aircrafts, and drones in urban low-altitude and ultra-low-altitude airspace. However, the lack of security algorithms in ADS-B makes it vulnerable to eavesdropping, jamming and spoofing attacks [44]. Wide-Area Multilateration (WAM), a derivative of Multilateration (MLAT), employs ADS-B to create a wide-ranging aerial surveillance network [45]. As useful as it is, WAM still suffers from security holes that stem from its underlying technology. Even if the ADS-B information changes, WAM can determine the aircraft position considering only the time of reception [46]. WAM functionality is not affected by ADS-B data tampering but is vulnerable to GPS spoofing attacks.

D. Ground network

To enable real-time sharing of data and TBO operations, the air traffic management (ATM) system uses the proven Transmission Control Protocol/Internet Protocol (TCP/IP) model. Within this framework, a system-wide information management (SWIM) architecture based on the TCP/IP model emerged [47]. The system introduces the concept of virtual information base as a link of aviation-related information exchange, so as to realize data sharing. But because of this, the SWIM architecture inherits the potential vulnerabilities and shortcomings unique to IP networks. More vulnerable to denial of service (DOS) attacks, data breaches, tampering, and spoofing attacks. Since its centralized architecture covers the entire civil aviation sector, any malicious attack on data within SWIM could pose a serious threat to the entire civil aviation sector. The vast scope and importance of SWIM in aviation underscores the seriousness of potential breaches and attacks on its data integrity and security.

V. AI BASED MITIGATION

In the context of the ATM system, cybercriminals employ a myriad of techniques to pose threats. Leveraging Machine Learning for mitigation purposes offers more efficient avenues for addressing these challenges.

A typical example is support vector machine (SVM), which is able to identify the error difference between GPS and UAS inertial navigation systems to detect GPS spoofing attacks on unmanned aerial systems (UAS) [48]. By constructing decision boundaries based on normal aircraft data, the method can effectively identify spoofing attacks without requiring additional hardware. Manesh [49] uses artificial neural network (ANN) algorithm to classify

legitimate signals and damaged signals, so as to determine whether GPS signals are attacked. The trained model shows high detection accuracy, low false positive rate, and wide adaptability, making it deployable on drones of various sizes. Demertzis [50] developed a framework for network flow forensics using ensemble learning algorithms. The framework can classify malicious activities, analyse malware traffic, and identify botnets, so the framework can effectively detect Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks. Also of note are generative adversarial networks (GANs) based on unsupervised learning (UL) algorithms, which play a key role in shaping secure and adversarial systems. GAN is good at generating realistic adversarial data, which enriches the ability of the system to generalize unforeseen adversarial attacks [51]. DoS-WGAN is a comprehensive framework utilizing Wasserstein Generative Adversarial Networks (WGAN) and gradient penalty techniques [52]. Network Intrusion Detection Systems (NIDS) are trained on Convolutional Neural Networks (CNN) or other supervised and deep learning algorithms. By autonomously simulating network attacks, DoS-WGAN breeds a more powerful NIDS capable of identifying and mitigating various adversarial scenarios. Autonomous driving technology based on reinforcement learning (RL) algorithms is booming. Its autonomy and anti-jamming capabilities are ideal for emergency hosting of the system. For example, remotely controlled drones can fly autonomously in the face of signal interception and system hijacking. Johansson [53] developed a policy model for UAS path planning using policy network algorithms. This approach addresses the challenge of automatically mapping the flight path of a long-range surveillance aircraft in the event of a loss of control due to jamming.

While ML algorithms offer strong solutions, they face practical challenges: adapting to environments, data availability, and the opacity of black-box models. ML seems versatile, but it requires significant pre-training for deployment, limiting its applicability. Deviating from training scenarios reduces accuracy [54]. ML's utility is wide, but its applications are constrained. For example, Manesh's GPS spoofing detection needs distinct training for different environments. Training data heavily affects ML-based systems [55]. They rely on data-driven learning, leading to biased models. UTM-based systems often use aviation or simulated data, with limitations. ML's black-box nature poses challenges. ANN and deep learning's complexity makes them hard to understand [56]. Despite efforts for interpretability, many algorithms lack explanation, complicating understanding, and development.

VI. CONCLUSION

This article examined current and emerging threats in aviation, including UAS Traffic Management (UTM), and discussed potential ways to address them. It categorized threat agents based on their goals and capabilities and analysed various cyber threat methods. The article also explored potential cyber-attack targets in ATM and UTM systems, along with traditional and AI-based defence methods. It's predicted that as aviation systems adopt more digital technology, they will face increased cybersecurity risks, especially in network security. Challenges include the complexity brought by AI technology and the normalization of AI-driven attacks and defences. To counter these challenges, there's a need to develop highly effective attack

detection technology with minimal false positives and missed detections.

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